

BUYING FLAT-SCREEN TVs MADE EASY!

If you are shopping for a small HDTV, expect to pay 20 to 25 per cent premium for a 1080p TV set over a 720p set. All else being equal, a 1080p model will better match much of the content you can now get from streaming satellite services and the native resolution of a Blu-ray disk player



■ SWATI SHARMA

Catching on a few minutes of entertainment on your television is a relief after a hectic day. TVs are making this experience even more exciting with advances in display technology. Video displays of TVs have evolved from cathode-ray tube (CRT) to flat-panel plasma, liquid-crystal display (LCD), LED and even three-dimensional (3D) displays. All these technologies have their own pros and cons.

The biggest drawback of CRT TV sets is their bulkiness and weight. Flat-panel plasma, LCD and LED TVs, on the other hand, can be as little as 2.5cm (1-inch) thick and hung

like wall paintings or put over a pedestal.

Plasma TVs do away with the bulky picture tube and electron beam scanning of CRT TVs and instead employ the burning of phosphors to generate an image. However, these still suffer from some of the drawbacks of CRT TVs, such as heat generation and screen-burn of static images. This problem has been solved to a great extent with the incorporation of pixel orbiting and related technologies.

Plasma TVs also have potentially shorter lifespan of the display. Earlier, it was 30,000 hours or eight hours of viewing a day for nine years, which was shorter than LCDs'. But with improvements in technology, it has

now increased with some sets rated as high as 100,000 hours.

There are certain advantages as well. Compared to an LCD TV, a plasma TV provides:

1. Better contrast ratio and deeper blacks
2. Better colour accuracy and saturation
3. Better motion tracking (little or no motion lag in fast-moving images)

In LCD TVs, fast-moving objects may exhibit lag artifacts. However, this is improving with the implementation of 120Hz and 240Hz screen refresh rates in higher-end models.

Since there are no phosphors to light up, less power is needed to operate an LCD TV. Its light source too generates lesser heat than a plasma or CRT TV's. Also, because of the nature of LCD technology, there is no radiation emitted from the screen itself. Increased image brightness over plasma makes LCD TVs better for viewing in brightly lit rooms.

Although LCD televisions do not suffer from burn-in susceptibility, individual pixels on an LCD television can burn out, causing small, visible, black or white dots to appear on the screen. Individual pixels cannot be repaired. The whole screen would need to be replaced if the individual pixel burnout becomes annoying to you.

Earlier, LCD TVs came mostly in smaller sizes. One reason was that quality-control issues limited the display size. Increasing the size of an LCD panel meant adding pixels and